

What is an NBA player worth?

Pricing wins, aging and injury risk with free data, 2010-11 to 2025-26

Viktor Sadowski · October 2026 · code: nba-moneyball

Summary

This paper prices every NBA player in wins and compares that with what he is paid. It uses only free public data: 16 seasons of play-by-play split into 577,807 lineup stints, two injury sources and salary and contract data. Player impact comes from regularized adjusted plus-minus (RAPM) with a box-score prior. It is projected forward with an aging curve and an injury-risk model, turned into wins above replacement (WAR) by calibrating on team results, and priced at what the market paid per win: about \$7.7M in 2025-26. The model predicts team wins from the previous season's player values with a correlation of 0.79. A second model prices players the way the market does, from 2,635 veteran contracts, and 415 trades show what the league pays for a draft pick. A season simulator turns rosters into playoff and title odds, and its 2026-27 forecast was frozen before opening night.

Six findings matter for a front office. Players peak at 28 and offense fades faster than defense. Injury proneness is real and predictable, but players who come back are as good as before, with one exception: after an ACL tear the first 20 games back are about 0.5 to 0.8 points per 100 possessions worse. The biggest surpluses sit on young players on rookie or early extension deals, while only 7 of the 27 players paid \$45M or more project to earn their salary next season. Three-point volume went from underpaid to overpaid around 2019-20, but what the market really overpays is scoring, and accurate shooters are the bargain. Defense does not win championships: in the playoffs a point of defensive edge is worth no more than a point of offensive edge. And when our value and the market's price disagree, ours predicts team wins better (error 6.4 wins per 82 vs 7.2), and the market's price adds nothing once ours is known. That gap is where a team can trade.

1. The idea

A team has a budget, the salary cap, and wants wins. So the useful question about a player is how many wins he adds per dollar, compared with what else the same money could buy. Baseball has priced players this way for decades with WAR. In basketball the pieces exist, plus-minus models on one side and public contract data on the other, but they are rarely put together into one price. This project does that, end to end, with data anyone can download.

The framework answers five questions:

- How many wins will a player add next season, and over the rest of his contract?
- How much of that is at risk because of age and injuries?
- What does a win cost on the market?
- Where does the market misprice players?
- What will other teams accept for him, and who is right when that differs from what he adds?

The goal is a blueprint: a pipeline a front office can rerun every summer, with clear inputs, a validation for each step and known limits. Three side questions run on the same data: do MVP voters see the same players as the model, are three-point shooters overpaid, and does defense win championships. A separate case study puts it all to work on one team: what trades give the 2026-27 Philadelphia 76ers the best shot at a title ("How do we get Bron his last ring?", in the same repository).

2. Data

Source	Content	Seasons
Play-by-play, stats.nba.com format (free mirror on GitHub)	every event incl. substitutions, regular season and playoffs	2010-11 to 2025-26
stats.nba.com via nba_api	game logs, minutes, ages, box scores, schedules	2010-11 to 2025-26
ProSportsTransactions injury log (pre-scraped CSV)	injury list moves with a note on the injury	2010-11 to 2019-20
Official NBA injury reports (PDF)	status and reason per player, every game day	2021-22 to 2025-26
Basketball-Reference	salary per player-season, current contracts, MVP votes	2010-11 to 2031-32
stats.nba.com via nba_api	draft history, 2026-27 rosters	2010 to 2026
Basketball-Reference	transactions: every trade with players, picks and cash	2013-14 to 2026-27

Table 1. Data sources. 2020-21 has no injury data and is skipped wherever an injury count is needed.

Lineups are rebuilt from the substitutions. Every game becomes a sequence of stints, stretches where the same ten players are on the floor, with points and possessions for both sides. Between 97.9% and 99.9% of each season's games pass the consistency checks. That gives 18,996 regular-season games and 1,339 playoff games. Salaries were matched to player ids by name and team, contracts by name over the last three seasons.

3. Method

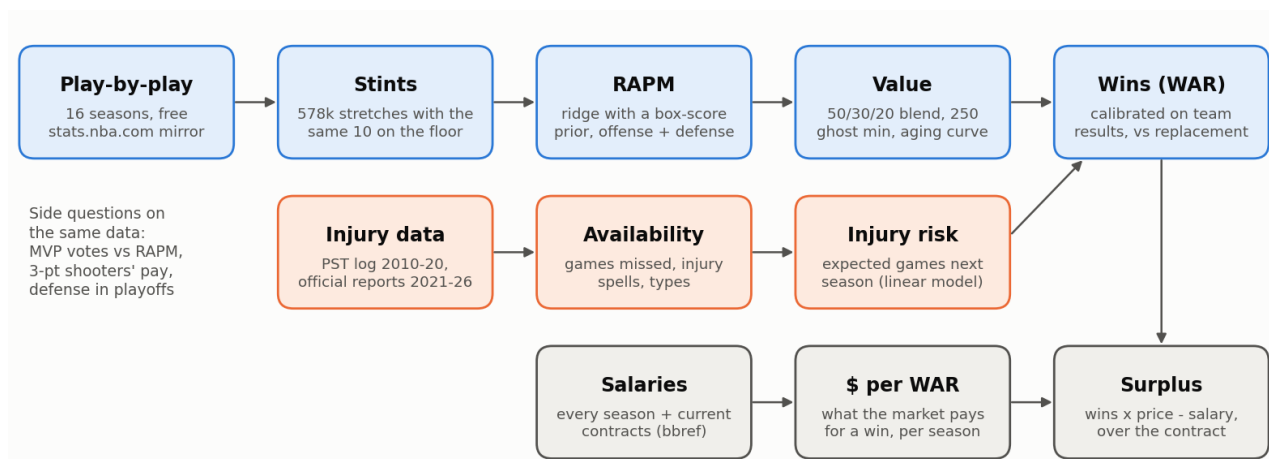


Figure 1. The pipeline. Blue: how good a player is. Orange: how often he plays. Grey: what he costs. Every box is one script in the code.

3.1 Impact per possession (RAPM)

For each season a ridge regression explains points per 100 possessions in every stint with the ten players on the floor, offense and defense as separate columns, weighted by possessions. The penalty is chosen by cross-validation with whole games held out. The result is O-RAPM and D-RAPM: points per 100 possessions a player adds on offense and saves on defense, compared with an average player.

A plain ridge shrinks every player towards 0, the league average. Players with few minutes, or who almost always share the floor with the same teammates, end up close to 0 whatever they did. So the ridge shrinks towards a box-score prior instead: a weighted linear model that predicts O-RAPM and D-RAPM from box stats per 100 possessions (points, shot attempts, free throws, threes, assists, turnovers, rebounds, steals, blocks, fouls and minutes per game). For each season the prior is fitted on the other seasons only, and the penalty is picked by cross-validation again (3,000 to 10,000). In a forward test, player values from one season predicting every stint of the next, the prior version beats plain RAPM in all 15 seasons and explains 44% more of the predictable variance.

3.2 Projection and aging

A single season of RAPM is noisy, so next season's value is a blend of the last three: weights 50/30/20, each multiplied by minutes played, plus 250 minutes at league average to pull small samples towards the middle. Each past season is first moved along the aging curve to the age the player will be next season. The aging curve is a fixed-effects regression (player talent plus one term per age) on RAPM, checked against the delta method and smoothed with a quadratic. In a backtest the age adjustment lowers the error of the next-season projection from 1.667 to 1.613 points per 100 (RMSE, minutes weighted). Without the prior the backtest preferred 1,000 ghost minutes; with it, much less extra pull is needed.

Rookies have no RAPM, and young players improve faster than the aging curve says. So a rookie starts at what his draft slot has been worth (value and minutes as a function of log pick, from every pick since 2010), and players in seasons 2 to 4 get a bump by pick group, learned from how far earlier projections fell short for the same group: about +1 point per 100 in year 2 for top-5 picks, 0.1 to 0.5 elsewhere. Both are tested out of sample (one draft left out, the bump from earlier seasons only). On the seasons from 2016-17 on, the first ones are needed to learn the bump, they bring the error from 1.546 to 1.527 for all players and from 1.642 to 1.585 for players in their first four seasons.

3.3 From points to wins

The blended value is shrunk, so it is put back on a real scale by calibrating against the next season's team point differentials: calibrated value = $1.19 \times \text{value} - 0.78$. Replacement level is what teams actually got from near-minimum contracts (the bottom 20% of salaries): -1.05 points per 100. One win equals 33.6 points of season point differential. Then

$$\text{WAR} = (\text{value} - \text{replacement}) \times \text{possessions} / 100 / 33.6$$

As a check, each team's wins are predicted from the previous season's player values and this season's minutes: correlation 0.79, mean absolute error 6.1 wins per 82 games (0.78 and 6.2 without the prior). A team of replacement players wins about 29 games.

3.4 Availability and injury risk

Games lost to injury come from the injury log up to 2019-20 and from the official reports after that. The two sources differ in how many short absences they catch, so injury games are normalized within each season and converted back with the official-report average (13.7 games per 82). A linear model on the last three seasons of injuries, age and minutes predicts next season's games missed. In a rolling backtest it beat both the league average and gradient boosting (mean absolute error 12.4, 12.9 and 12.5 games). Projected minutes are expected games times minutes per game. Players who ended the season still injured get the rest of their absence on top: from all earlier absences of the same type that had already lasted as long, and of the same severity (tear, fracture or surgery or not), how much of next season they still missed. The comparison is in calendar days, so the summer counts as recovery time. It uses established players who came back with the same team (a return after changing teams mixes injury and free agency) and leaves out absences over the 2011 lockout and the 2020 break. They also carry the rust of their first 20 games back.

3.5 The price of a win, and surplus

The market price of a win in a season is the total salary paid above the near-minimum level, divided by the total positive projected WAR it bought. It was \$7.7M in 2025-26 and grows with league payroll, 9.1% a year over the last ten seasons. Later contract years move the player further along the aging curve with the same availability, and WAR is floored at 0 since a team can always bench a player. Surplus over a contract is the sum over its years of $\text{WAR} \times \text{price per win} - \text{salary}$. Option years are valued as options: the team keeps a team-option year only if the player is worth his salary, so it is worth $E[\max(\text{worth} - \text{salary}, 0)]$; a player leaves on a player option when he is underpaid, so the team's side is $E[\min(\text{worth} - \text{salary}, 0)]$. The spread of the projection 2 to 5 years out is measured on our own history, and non-guaranteed years count as team options. Rosters come from nba.com: a contract on a team whose roster doesn't have the player is dead money, it counts for the payroll but not on the court.

3.6 The market's price

Surplus says what a player adds. A trade also needs what other teams will accept, so a second model prices players the way the market does. It takes every new veteran contract since 2013-14 (2,635, found where a salary jumps outside the raises one deal allows) and fits the first-year salary as a share of the cap on what the player had done before signing: points, rebounds, assists, minutes and games, last season and the three before, plus shooting, age and our RAPM value. Minimum and max deals only say "at most" and "at least", so the fit is a Tobit. With the season left out it misses by 3.1% of the cap on deals between the minimum and the max (about \$5M a year, r 0.71). A player's market value is that price for every season left on his contract minus his salary.

Draft picks are priced two ways. What a pick turns into: the market value of what drafted players did over their four rookie-scale seasons minus their salaries, by slot. And what trades pay for one: with players at market value, the gap between the two sides of each of 415 trades since 2013-14 is what the picks and cash had to make up, solved by least squares with a bootstrap over trades.

3.7 From rosters to seasons

A team's strength is the sum of its players' values per 100 possessions, minutes given to the best players first and the rest filled at replacement level. Two corrections sit on top. Lineup shape: every player gets a role from his box score (rebounds, blocks, assists and threes per 36, clustered into guards, wings and bigs), and 467,485 stints show what a lineup with too few bigs or guards costs beyond the players' RAPM. Playoff rotations: playoff minutes go more to the top of the roster, and the playoff rounds use those shares. The simulator then plays 10,000 seasons with the real schedule format, the play-in and best-of-seven series.

4. Results

4.1 Who is valuable

Player	Age	Value per 100	Expected injury games	WAR	WAR if healthy	Lost to injury risk
Shai Gilgeous-Alexander	28	9.1	12.1	13.3	15.8	2.5
Nikola Jokić	32	7.8	13.1	11.4	14.0	2.5
Victor Wembanyama	23	7.1	14.4	9.8	11.9	2.2
Luka Dončić	28	5.7	14.7	9.1	11.3	2.2
Amen Thompson	24	4.5	11.1	7.5	8.7	1.2
Kon Knueppel	21	4.3	9.0	7.5	8.4	0.9
Giannis Antetokounmpo	32	5.5	19.4	7.1	9.6	2.5
Donovan Mitchell	30	4.7	13.4	7.0	8.6	1.6
Tyrese Maxey	26	3.7	13.9	7.0	8.5	1.5
Kawhi Leonard	36	5.3	17.2	6.8	9.0	2.2

Table 2. Highest projected WAR for 2026-27. Value is the calibrated, age-adjusted value in points per 100 possessions above average, including the young-player bump. Age is the age at the end of the season, the way stats.nba.com counts it, so players with a spring birthday are a year older here than in most lists.

The model and the MVP voters mostly agree on who the best players are. The median MVP winner since 2010-11 ranks second in that season's RAPM, 75% of winners were in the RAPM top five and half were first. They disagree on why: vote share correlates with offensive RAPM (Spearman 0.48) and barely with defensive RAPM (0.10, not significant). Voters reward what they can see on the box score.

4.2 Aging

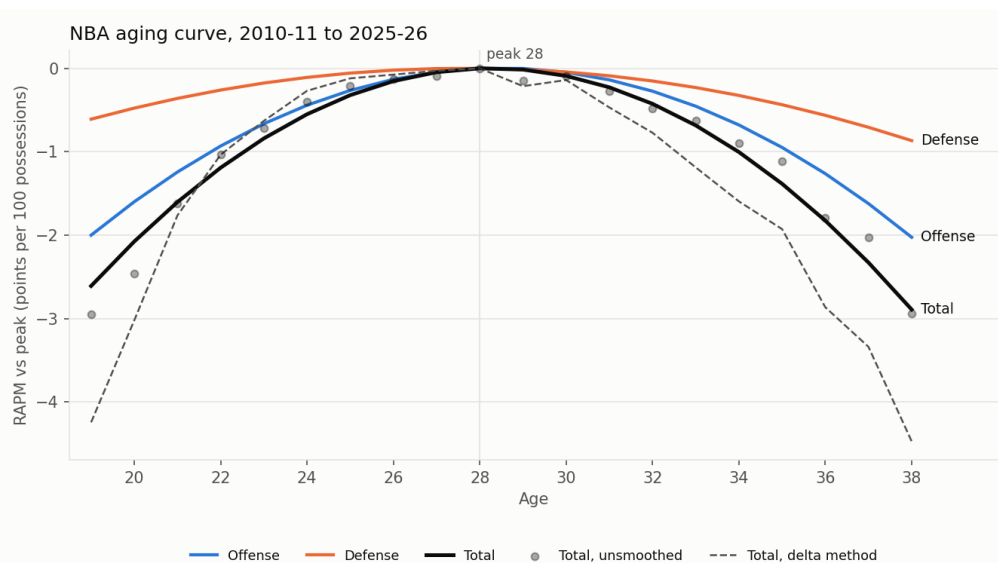


Figure 2. Aging curve, points per 100 possessions compared with the peak. Fixed effects (lines, smoothed) and the delta method (dashed) agree up to about 33.

Players peak at 28 and stay within 0.1 points of the peak from 27 to 30. A player is 0.8 points per 100 below his peak at 23, 0.7 at 33, and 1.4 at 35. The decline after 30 is slower than the rise before 25. Offense drives both the rise and the fall; defense moves about a third as much. In money: a four-year deal signed at 30 covers exactly the years where the curve gets steep.

4.3 Injuries

Injury history predicts more injuries, but not a worse player when he's on the floor

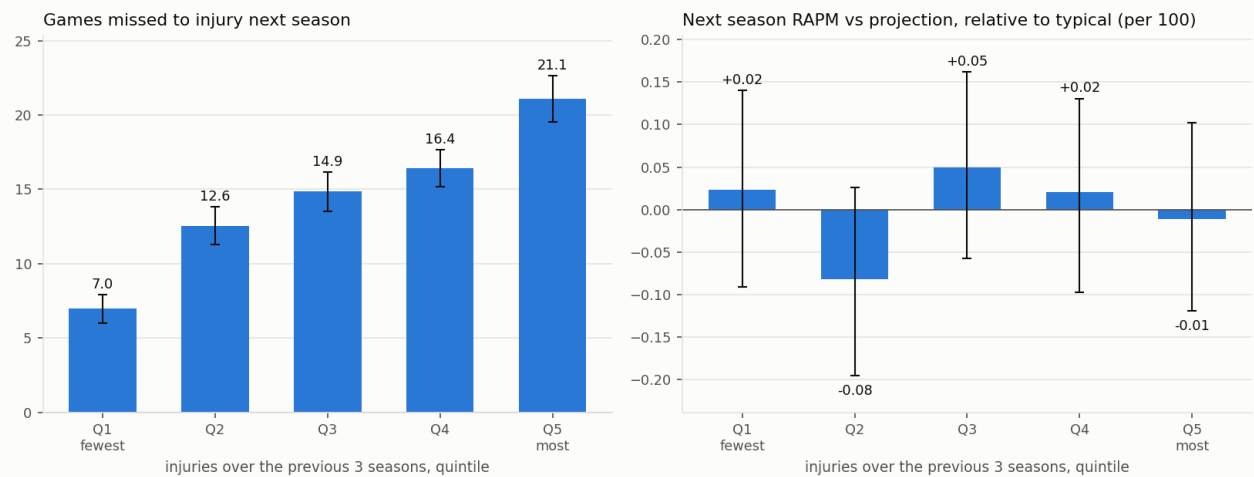


Figure 3. Left: games missed to injury next season, by injury history over the last three seasons. Right: next season RAPM compared with the projection, by the same quintiles. 95% bootstrap intervals.

Injury proneness is real. Games lost to injury correlate from one season to the next (Spearman 0.33), and 16% of the variation in injury rate belongs to the player. Players in the bottom fifth of injury history miss 7.0 games the next season, the top fifth 21.1. What injury history does not predict is decline: next season's RAPM compared with the projection is flat across all five groups. The discount for an injury-prone player belongs on his expected games, and the talent estimate can stay as it is.

A season is a clumsy unit for the question of what an injury does. A player hurt in October is back by January and his next season is really his second season back, while a player hurt in March starts the next season straight out of rehab. So each absence of 10 or more games is also measured in games played around it: his last 82 games before the injury as the baseline, and his first 20 and first 82 games after he is back. Impact in each window comes from one ridge over all 16 seasons, with the post-injury windows fitted as a change from the same player's baseline. The same windows around random points in healthy stretches serve as the control group. That gives 1,295 injuries and 2,544 control stretches.

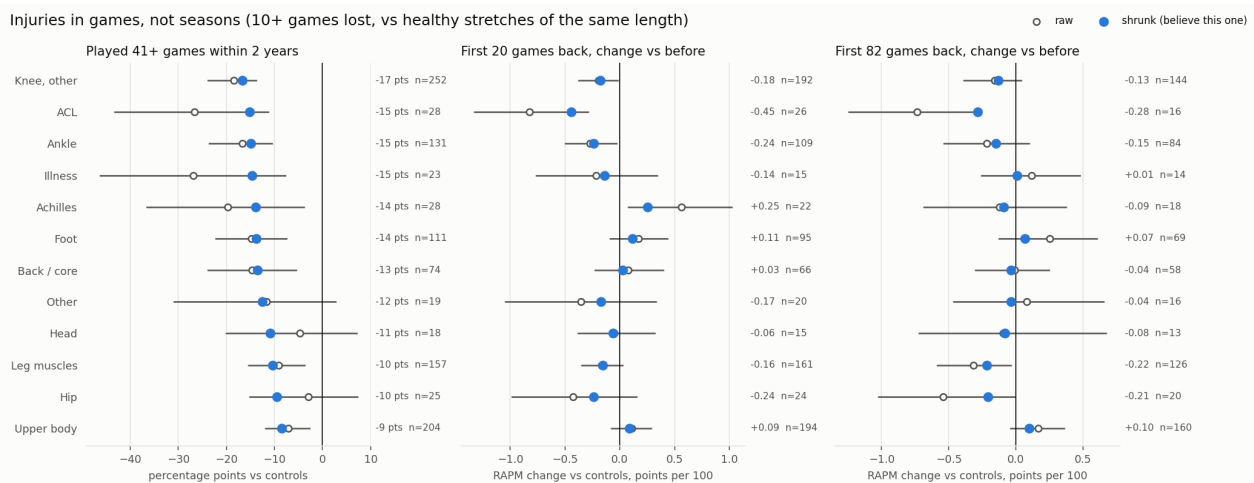


Figure 4. Injuries measured in games. Left: share who play 41+ games within two years, compared with the controls. Middle and right: RAPM change from before the injury, compared with the controls, over the first 20 and the first 82 games back. Blue is the estimate after empirical-Bayes shrinkage towards the average type, which is the one to believe for small groups.

Getting back on the floor depends on how long the absence was: 92% of the controls play 41 games within two years, 7 points fewer after 10-19 games out and 58 points fewer after a full season out. Once the noise is removed, injury types differ by only about 4 points. Knee injuries other than the ACL are the worst group (-17).

The ACL is the one injury with clear rust: the first 20 games back are 0.83 points per 100 worse (interval -1.33 to -0.29 , 0.45 after shrinkage), and over the first 82 games most of it is gone. Other types stay within about ± 0.25 points. The one exception, $+0.25$ for 22 Achilles returners, mostly reflects the aging correction for older players who sat out a full year.

In the projections the injury risk costs about 22% of the league's healthy WAR. For the 65 players worth 4+ WAR when healthy it is 1.3 WAR a season on average, 21% of their value. The largest shares belong to players out with a long injury or just back from one: Jimmy Butler (71%), Kyrie Irving (44%), Damian Lillard (43%), Fred VanVleet (41%) and Tyrese Haliburton (36%). 54 players were still listed out at the end of the season and 42 more had played fewer than 20 games since coming back.

4.4 Surplus: who is worth his contract

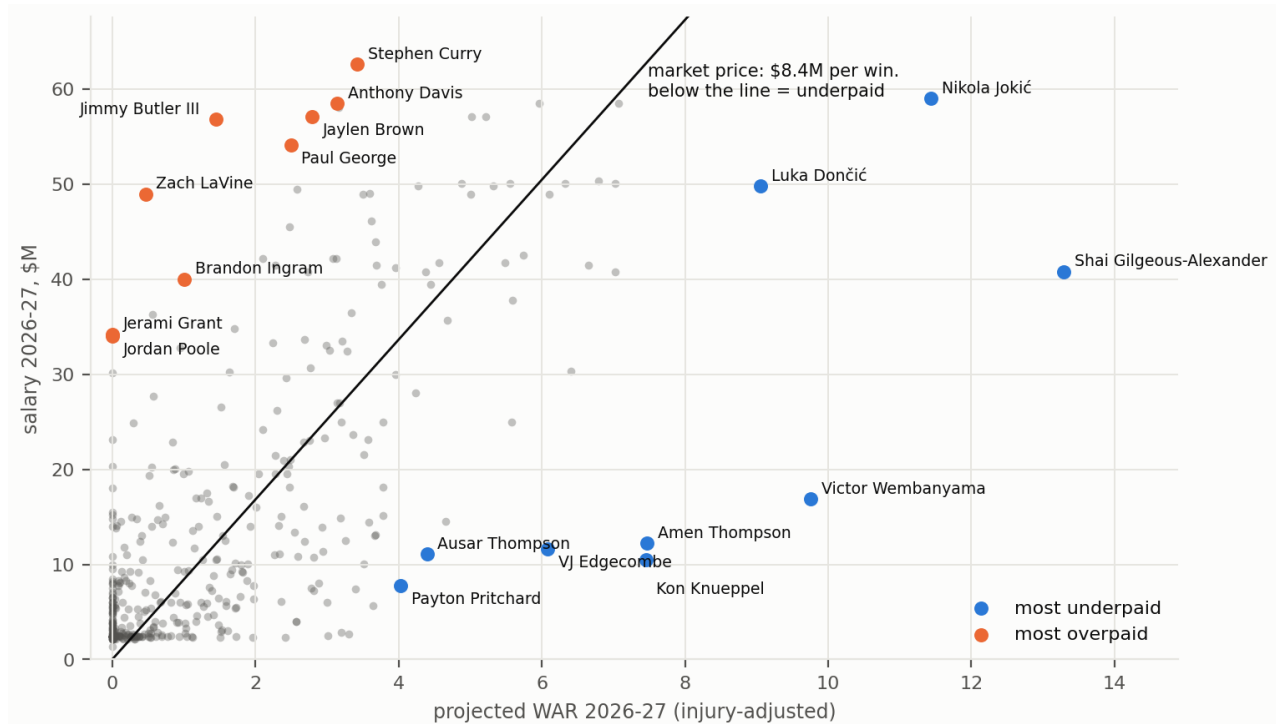


Figure 5. Projected WAR against salary for 2026-27, every player with a contract. The line is the market price of a win. Players below the line produce more wins than they are paid for. Highlighted: the nine largest surpluses and deficits for next season.

Most underpaid	Age	Years	Salary \$M	WAR	Surplus \$M	Most overpaid	Age	Years	Salary \$M	WAR	Surplus \$M
Wembanyama	23	6	269	63.0	+313	Trae Young	28	4	213	10.3	-114
Amen Thompson	24	6	220	48.4	+295	Jaylen Brown	30	3	183	7.6	-113
Gilgeous-Alexander	28	5	314	65.3	+263	Keyonte George	23	6	162	4.3	-113
Kon Knueppel	21	3	36	24.3	+189	Joel Embiid	33	3	188	8.3	-112
VJ Edgecombe	21	3	39	20.4	+149	Paolo Banchero	24	5	241	14.2	-99
Dyson Daniels	24	4	100	24.1	+133	Ayo Dosunmu	27	5	112	2.4	-85
Chet Holmgren	25	5	241	34.7	+110	Dillon Brooks	31	4	93	1.7	-78
Neemias Queta	27	5	59	16.4	+106	Devin Booker	30	4	251	18.4	-77

Table 3. Surplus over the whole remaining contract (salary, WAR and surplus summed over its years), with team and player options valued as options.

The market price of a win for 2026-27 is about \$8.4M. At that price only 37% of players under contract project to earn their salary next season, and 7 of the 27 players paid \$45M or more. The largest surpluses in the league sit on young players on rookie-scale or early extension deals, and on players whose value comes from defense and efficiency: Dyson Daniels and Neemias Queta return two to three times their salary. The largest deficits sit

on high-usage scorers whose on-court impact is modest by RAPM, and on stars past 32 on the last years of big deals.

Options move the picture for a handful of contracts. A player option costs the team exactly in the good outcomes: Wembanyama's and Gilgeous-Alexander's final-year player options take \$88M and \$80M off their surplus, Jokić's \$39M, because a player who is worth more than his salary leaves. Team options work the other way and help most on young, uncertain players: the option years on rookie deals like Ace Bailey's or Jeremiah Fears' are worth about \$16-19M to their teams, since a bad outcome can simply be declined. Across the league, options and non-guaranteed years add up to a net \$160M for the teams.

4.5 Are three-point shooters overpaid?

For every player-season from 2011-12 on, salary is expressed in wins at that season's price and regressed on the projected WAR made the summer before, 3-point attempts per 36 minutes and 3-point percentage from the previous season (both as z-scores within the season), a rookie-contract flag and age. A positive coefficient on shooting means shooters are paid more than the wins they are expected to bring.

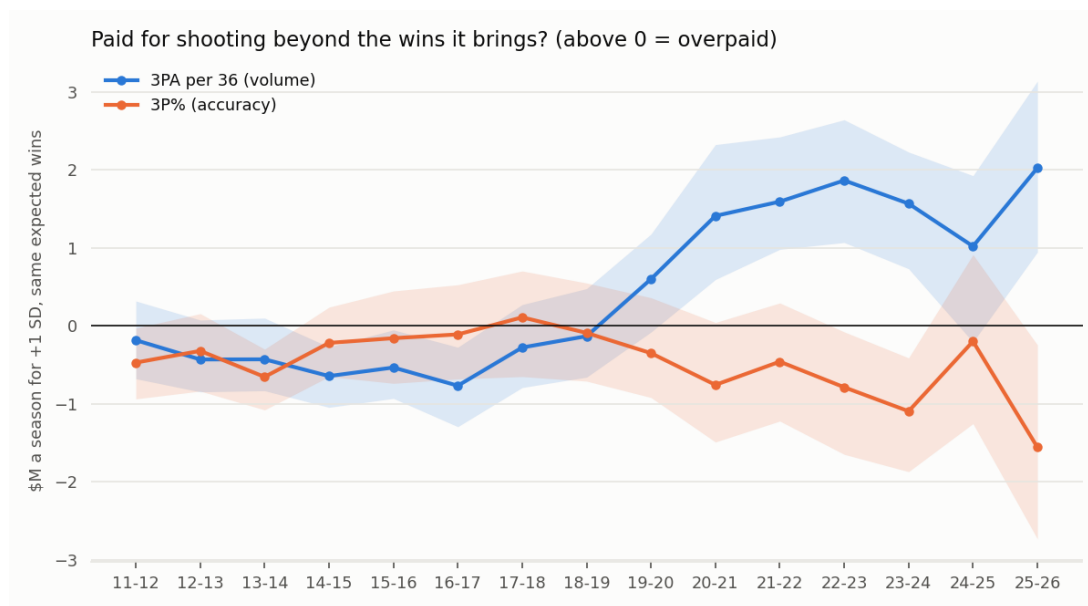


Figure 6. Extra salary per season for one standard deviation more 3-point volume (blue) or accuracy (orange), at the same expected wins. Bands are 95% bootstrap intervals over players.

\$M per season, +1 SD, same expected wins	2011-12 to 2014-15	2015-16 to 2019-20	2020-21 to 2025-26
3PA per 36	-0.43 [-0.8, -0.1]	-0.28 [-0.6, 0.0]	+1.57 [1.0, 2.0]
3P%	-0.43 [-0.7, -0.1]	-0.14 [-0.4, 0.2]	-0.79 [-1.2, -0.4]
3PA per 36, with points per 36 as control	-0.86 [-1.1, -0.6]	-0.58 [-0.9, -0.2]	+0.44 [-0.0, 0.9]
Points per 36 (same model)	+1.84 [1.4, 2.3]	+1.58 [1.0, 2.0]	+3.83 [3.3, 4.5]

Table 4. Pay for shooting beyond expected wins, by era, with 95% intervals.

Until 2019 high-volume shooters were slightly underpaid for the wins they brought. Around 2019-20 the market caught up and then passed them: since 2020-21 one standard deviation more volume comes with about \$1.6M a season extra at the same expected wins. Accuracy was never paid for, and since 2020-21 accurate shooters are paid about \$0.8M a season less than their wins are worth. Once points per 36 minutes is in the model most of the volume premium goes away (+\$0.4M, interval touching 0), and points carry a large premium of their own (\$3.8M per standard deviation). So the market pays for scoring volume, threes are one way to score, and the accurate shooter is the bargain.

4.6 Does defense win championships?

Does defense win championships?

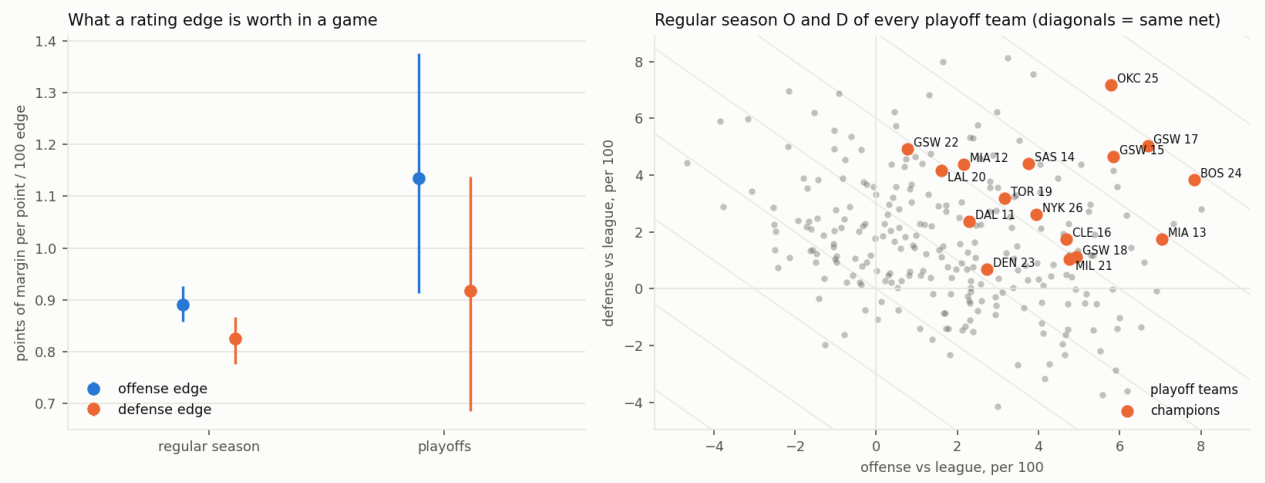


Figure 7. Left: points of game margin per point of regular-season rating edge, offense and defense, with 95% intervals (playoffs bootstrapped over series). Right: every playoff team since 2010-11; diagonals connect teams with the same net rating.

Each team's regular-season offense and defense are measured against the league average, and the margin of every game is regressed on the two teams' offensive and defensive edges. In the regular season a point of offensive edge is worth 0.89 points of margin and a point of defensive edge 0.83. In the playoffs the numbers are 1.14 and 0.92; the difference, -0.22 (interval -0.50 to $+0.05$), points slightly the other way. The same holds for wins. Champions are no more defensive than other good teams: 10 of 16 were top five in offense, 9 of 16 in defense, and 9 of 16 were top three in net rating. Net rating wins championships, and it does not matter much which end it comes from. If anything a defensive rating predicts a little less, probably because it holds more luck, such as the opponents' shooting.

4.7 Our price vs the market's: who is right?

The market model says what teams pay for a box score. Consistency is paid for: three seasons of stats predict a contract better than the last one, and games played over three years is one of the strongest terms. Volume comes first, impact a little after. So the two prices disagree, most on high-usage scorers on big deals. Which one is right can be tested with the one judge neither side controls: wins. For every team-season since 2013-14 the players were valued three ways before the season (payroll, the market model and our projected WAR), each scaled by how much they then played.

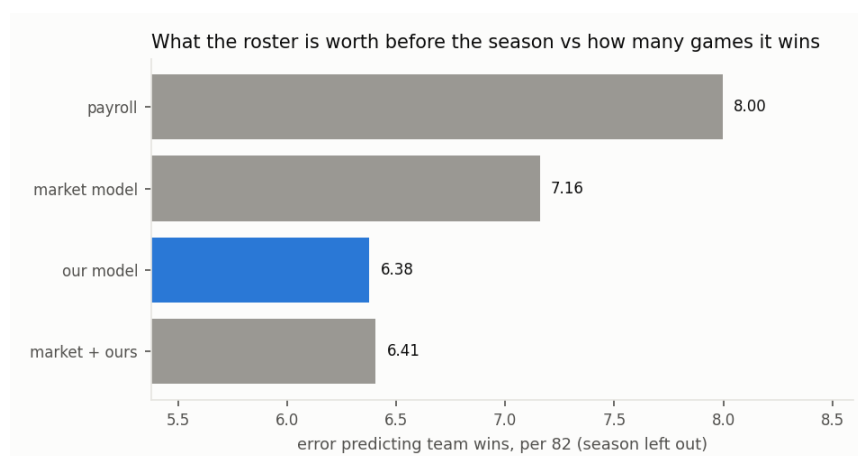


Figure 8. Error predicting team wins per 82 games from what the roster was worth before the season, 390 team-seasons, each season predicted from the others.

Payroll misses by 8.0 wins, the market model by 7.2 and our model by 6.4. In one regression with all three, ours carries it: +8.9 wins per standard deviation, the market model -0.7 (95% interval -2.7 to $+1.5$) and payroll $+1.7$ (probably what teams know about health that the stats don't). Once our value is known, the market's price says nothing more about wins. It is still the right guide to what another team will accept: trades balance far better in the market's money than in ours (average miss 5.5% of the cap vs 9.7%). One caveat: our aging curve and box-score prior are fitted on all seasons, later ones included, while the market model is fitted with the season left out.

The same trades price draft picks. A future first buys about \$5.3M of market value (95% interval $-\$0.4$ M to $\$11.3$ M) and a second \$1.4M ($\0.2 M to $\$2.9$ M), while the average first turned into \$17.9M of value over its rookie deal and the average second \$3.3M. The slot curve has the shape of the published research: #10 is worth about half of #1 and #30 a fifth. A traded pick buys less than it turns into, so picks close gaps in a trade but rarely buy a star.

4.8 The 2026-27 season

Lineup shape matters mostly at the guard spots. Compared with two bigs and two guards on the floor, no big costs 0.8 points per 100 (interval -1.4 to -0.1), one big is as good as two, no guard costs 2.1 and one guard 0.7, all on top of the players' RAPM. For 2026-27 that moves teams from -0.8 (Houston, short on guards) to $+0.6$ (Milwaukee). Playoff rotations are real but small: the top five players go from 57% of the minutes to 61%, which adds $+0.2$ to $+0.8$ points per 100 in the playoffs, most for star-led Denver and Oklahoma City.

Team	Wins	Net per 100	Playoffs	Finals	Title
Oklahoma City	57.9	+8.1	99%	44%	32%
San Antonio	53.6	+5.7	97%	21%	14%
Boston	51.7	+4.7	94%	24%	11%
Houston	51.5	+4.9	95%	15%	9%
New York	48.4	+3.2	87%	15%	5%
Denver	48.1	+3.4	88%	9%	5%

Table 5. Projected 2026-27, 10,000 simulated seasons. The full forecast for all 30 teams and every player was frozen on October 1, before opening night (forecasts/2026-27-v2 in the repository), to be scored after the season.

5. Using it in a front office

5.1 Contract ceilings

Before a negotiation starts, the ceiling for a deal is the sum over its years of projected WAR times the price of a win, with the aging curve and the player's own injury projection built in. Anything below the ceiling is surplus for the team. Derrick White, for example, projects at 6.4 WAR next season, about \$54M of wins on a \$30M salary; at 32 the ceiling for each extra year falls, which the curve prices directly.

5.2 Trades and extensions

Surplus is the asset a team actually trades. Rookie-scale stars are the most valuable contracts in the league by a wide margin, and that surplus shrinks the moment an extension starts. Two players with the same reputation can differ by hundreds of millions in surplus, depending on age and years left.

A trade needs both prices: ours for what a player adds, the market's for what the other team will accept. The useful trades sell players the market still likes and we don't, and buy the reverse. The case study does this for Philadelphia. Joel Embiid and Jaylen Brown are priced close to their salaries by the market and about \$110M underwater over their contracts by our model. Three trades that are even or better for every partner at the market's price, legal under the 2026-27 CBA (salary matching by apron, trade bonuses, roster spots), take the 76ers from 46 to 58 projected wins and from 3.5% to 27% title odds while staying under the first apron and giving up only second-round picks.

5.3 Pricing injury risk

Injury history predicts missed games and does not predict a worse player. The protection should therefore target availability: games-played incentives, partial guarantees and shorter terms for the high-risk group, whose players are expected to miss three times as many games as the low-risk group. The model gives a per-player expected games number for every negotiation.

5.4 Return-to-play planning

After an ACL tear, plan for a weaker first 20 games, about half a point per 100 below the player's level, and set minutes and matchups accordingly. The first month back says little about the player or the deal. For other injury types there is no measurable rust once a player is cleared.

5.5 Where the market is mispriced

- Scoring volume is priced well above what it adds in wins. Paying for points per 36 is the most common overpay in the data.
- Three-point volume is priced in since about 2019-20. Accuracy is not: at the same expected wins, accurate shooters are paid about \$0.8M a season less per standard deviation.
- Defense-first and efficient role players deliver the largest surpluses outside rookie deals, and MVP voters ignore defense entirely, which suggests the wider market does too.
- Picks are cheap to buy: a traded first costs about a third of what the average first turns into.
- For the playoffs, build net rating the cheapest way. There is no premium for getting there with defense.

5.6 Running it

The pipeline reruns every summer from free data, one script per step (appendix B). Each step has its own check: cross-validation for RAPM, a backtest for the projection and the injury model, and the team-wins check for WAR. A team can plug in better inputs without touching the rest: tracking data or a box-score prior for RAPM, its own medical data for the injury model, and its cap sheet for the contract mechanics. Each season's forecast gets frozen before opening night so it can be scored honestly afterwards.

6. Limitations

- RAPM is noisy and cannot fully separate players who almost always share the floor. The box-score prior helps, but what the box score misses (screens, spacing, positioning on defense) still has to come from the lineup data. Tracking data would be the next step.
- The win scale depends on the calibration and the replacement level. A replacement team wins about 29 games here, and every WAR and dollar number moves with that choice.
- Contracts: options are valued on expected value, one year at a time, and ignore extensions signed instead of an opt-out. The trade search models the 2026-27 trade rules (salary matching by apron, trade bonuses, non-guaranteed salaries, roster spots), but not cap holds, exceptions for free agents or planning over several seasons. The price of a win is the market average, while a contender may rationally pay more at the margin.
- The market model only sees new veteran contracts and the box score. It doesn't know about health records, agents or timing, and pick prices from trades are noisy (the first-round interval includes 0).
- Lineup shape uses three box-score roles. It catches lineups without bigs or ball handlers, not finer fit like spacing or two players who need the ball.
- Injury data: two sources with different completeness, normalized within season. No data for 2020-21. Injuries in the playoffs show up at the next season's first report. Some injury groups are small (16 to 28 ACL cases per measure).
- The salary analyses show associations: timing of signings and contract years are not modeled, and the rookie-contract flag is approximate.

- Model results will disagree with reputation for some players, most visibly high-usage scorers. That is partly the point, and partly the noise in the first bullet.

Appendix A. Technical details

Step	Choice	Validation
Lineups	starters inferred per period, sub names matched to ids with aliases and suffix rules; score fixes for rows that reset to 0-0	97.9-99.9% of games kept per season
RAPM	ridge, 2 rows per stint (each side on offense), possession weights, home-court term; penalty by 5-fold GroupKFold over games	CV error vs league-average baseline per season
Box prior	weighted linear fit, box stats per 100 (300 ghost possessions) + minutes per game -> O and D; fitted without the season itself and the next; ridge on $y - X \cdot \text{prior}$	forward stint test: better than plain in 15 of 15 seasons, +44% variance explained
Value	50/30/20 $\times$ minutes, 250 ghost minutes at 0, each season aged to next season's age	backtest RMSE 1.667 to 1.613 with aging
Rookies and young players	draft slot -> rookie value and minutes (log pick), bump for seasons 2-4 by pick group learned from earlier seasons	backtest RMSE 1.546 -> 1.527, seasons 1-4 1.642 -> 1.585
Aging	fixed effects (player + age dummies, ridge $\alpha=1$), delta method as check, weighted quadratic	peak 28; methods agree to about 33
WAR	calibration $1.19 \times \text{value} - 0.78$ on next-season team differentials; replacement -1.05 from bottom-20% salaries; 33.6 points per win	team wins $r = 0.79$, MAE 6.1
Injury risk	normalized injury share, 3-year blend, age, minutes; linear, shrunk to the mean	rolling backtest MAE 12.36 vs 12.88 (league average) vs 12.46 (boosting)
Injury windows	10+ games out; pre = last 82 games (after the previous return), post = first 20 / 82 games back; post windows fitted as change from pre (offset); controls at pivots in healthy stretches; separate fits for injuries and controls	1,295 injuries, 2,544 controls; empirical-Bayes shrinkage, bootstrap intervals
Current injuries	still out at season end (10+ games or a tear/fracture/surgery note); remaining games from earlier absences of the same type that lasted as long (established players only); rust for the first 20 games	54 still out, 42 back fewer than 20 games
Options	team option $E[\max(W - S, 0)]$, player option $E[\min(W - S, 0)]$, W normal with $sd = a + b \cdot \text{WAR}$ by horizon, fitted on past projections; next season's options count as decided	$sd\ 0.65 + 0.15 \cdot \text{WAR}$ (2 years out) to $1.11 + 0.38 \cdot \text{WAR}$ (5 years)
Price of a win	$\Sigma(\text{salary} - 20\text{th percentile}) / \Sigma \text{positive projected WAR, per season}$; 9.1% yearly growth	\$7.7M in 2025-26
Threes	salary in WAR units $\sim$ projected WAR + 3PA/36 + 3P% (shrunk to 35% with 100 attempts) + rookie flag + age + age ²	player bootstrap, by era and season
Defense	game margin $\sim$ O edge + D edge, regular-season ratings (leave-one-game-out in the regular season)	series bootstrap for playoffs; logit on wins as check
Market value	Tobit on first-year salary share of new veteran contracts (censored at 3% and 24.5% of the cap), box stats last season + 3 seasons, age, RAPM value	season left out: MAE 3.1% of cap, $r\ 0.71$
Pick prices	value in - out = $p1$ (1sts out - in) + $p2$ (2nds out - in) + cash + swaps, least squares, bootstrap over trades	415 trades, 2013-14 to 2026-27
Who is right	team wins $\sim$ payroll / market price / our WAR of the roster before the season, each scaled by minutes played vs expected	leave one season out, 390 team-seasons
Season simulator	net per 100 from rosters (best players first, filler at replacement) + lineup shape; playoff minute shares in the playoff rounds; real schedule format, play-in, best-of-seven	team wins MAE 6.1; playoff game RMSE 13.913 -> 13.901
Positions	roles from box stats per 36 (rebounds, blocks, assists, 3PA), k-means into guard / wing / big; stint margin minus the ten players' RAPM $\sim$ bigs and guards on each side	467,485 stints; bootstrap over games

Appendix B. Reproducing it

Everything runs from the repository with Python 3 (pandas, numpy, scipy, scikit-learn, matplotlib, pdfplumber). stats.nba.com blocks most cloud servers, so the roster and box-score pulls run from a normal machine. In order:

```
check_api, ingest, lineups, rapm, players, box_prior, aging, draft, value, mvp
injuries_pst, injury_reports (download, then --parse), scrape_salaries, rosters
availability, injury, injury_types, injury_windows
current_injuries, war, rookies, surplus (uses options), threes, playoffs, defense
market_value, scrape_transactions, trade_value, playoff_stints, playoff_rotation, positions
season_sim, arbitrage, freeze --version v2
report/make_figures.py, report/build_report.py
```

Data: play-by-play from github.com/shufinskiy/nba_data; stats.nba.com through nba_api; injury log from github.com/gboogy/nba-injury-data-scraper (ProSportsTransactions); injury reports from the NBA's official PDFs; salaries, contracts, transactions and MVP votes from basketball-reference.com; draft history and rosters from nba.com.